Lead 5b: the D_4-symmetrised S_5 as four weighted unknowns

2026-10-15, lane d4-s5. Study `research/d4_symmetrised_s5_20261015`. Protocol committed in `87fe78b63` before any measured run. Runs 06:38Z–07:43Z, one thread each, under a 2.5 GB cap, on a shared 4-vCPU container. Class: engineering, a constant of one phase. No end-to-end S.

The lead was already half-run

The encyclopedia lists the D_4-symmetrised S_5 as "never run". That entry is stale. Candidate C8 (PR #1785, `research/gaudry_quartic_d4_20261013`) ran it on 2026-10-13:

- form: five **unweighted** unknowns $(\sigma_1, \sigma_2, \sigma_3, \sigma_4, \pi)$ with $\pi^2 = \sigma_4$, Macaulay degree 14;
- matrix: 13412×11628;
- cost: $C_4 = 2.54 \cdot 10^{10}$, a factor of 48 below the plain S_4 form's $1.213 \cdot 10^{12}$. The projection was ≈ 630.

This study builds the **weighted** Macaulay solve that the GLV note §9.4 named and nobody built.

Construction (derived)

Curve and base. The curve is over $\mathbb{F}_{p^4}$ with $T = (x_0, 0)$ and $f'(x_0) = c^2$. The coordinate is $Y = \frac{x - x_0 - c}{x - x_0 + c}$, with $Y(P + T) = -Y(P)$. The base is $\{P : Y(P) \in \mathbb{F}_p\}$.

Group and invariants. $D_4 = (\mathbb{Z}/2)^3 \rtimes S_4$, of order 192, acts by permutations and by even sign changes, i.e. by even T-translates. The invariants are $\sigma_k = e_k(Y^2)$ and $\pi = \prod Y_i$.

System. Four Weil components in $(\sigma_1, \sigma_2, \sigma_3, \pi)$, each of weighted degree 8 under the weights $(2, 2, 2, 1)$.

Bézout. Two derivations agree:
- weighted: $8^4/(2 \cdot 2 \cdot 2 \cdot 1) = 512$;
- orbits: $4096/(192/24) = 512$.

Macaulay bound. $32 - 7 + 1 = 26$, at which the matrix has **4,200 columns**. C8 has 11,628 and the plain form 40,920.

Controls (measured)

- **Invariance detector.** A wrong c makes the invariant interpolation fail; the true c passes. Checked at $p = 13$ and $p = 269$.
- **Generic planted weighted systems.** 9 of 9 have quotient 512 and recover the planted root, at 1.39–$1.45 \cdot 10^9$.
- **Planted targets.** The planted quadruple is recovered by arm w in 41 of 42 and by C8 in 23 of 24.

Results (measured)

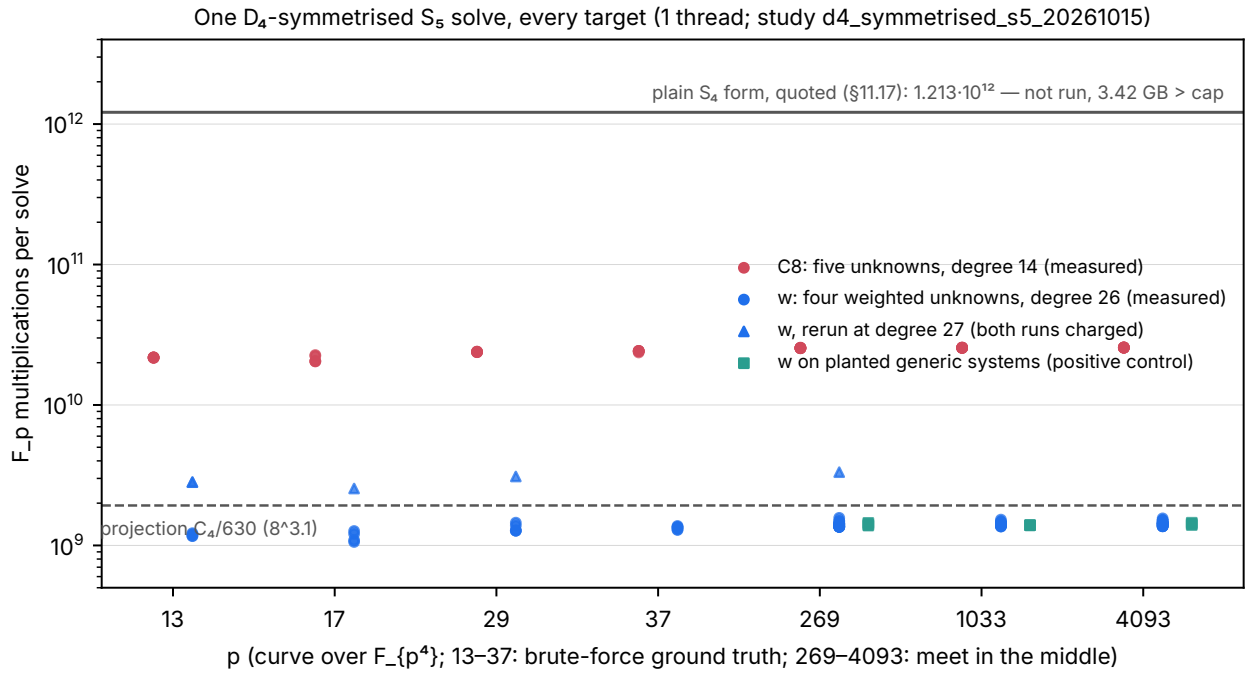


Figure 1: F_p multiplications per solve, every target. C8 (red) and the weighted form (blue) run on the same targets. The plain level is quoted from §11.17; it was not run, because its 3.42 GB matrix is over the cap.

p	C_w median	C_{c8} median	paired C_{c8}/C_w	$1.213 \cdot 10^{12}/C_w$	ground truth
13–37	1.21–$1.34 \cdot 10^9$	2.05–$2.42 \cdot 10^{10}$	17.9–18.5	905–1005	brute 23/24, MITM 23/24
269	$1.411 \cdot 10^9$	$2.542 \cdot 10^{10}$	18.0 (16.3–18.7)	860 (772–893)	MITM 20/20
1033	$1.406 \cdot 10^9$	$2.554 \cdot 10^{10}$	18.2 (17.4–18.7)	863 (823–888)	MITM 20/20
4093	$1.437 \cdot 10^9$	$2.561 \cdot 10^{10}$	17.8 (16.5–18.6)	844 (778–885)	MITM 20/20

- **Lift-back.** The split, F_p square roots, lifts and group checks of all 8 translates cost at most $2.8 \cdot 10^5$, which is at most $2 \cdot 10^{-4}$ of C_w.
- **Degree-27 reruns.** 5 of 101 solved targets had a degenerate eigenspace and reran at degree 27, charged at 2.5–$3.3 \cdot 10^9$ each.
- **One target unsolved.** At $p = 17$, planted target 4 has quotient 514 and is unsolved by both arms; brute force finds 4 decompositions there.
- **Exactness.** On every other target the two arms and the ground truth agree exactly.

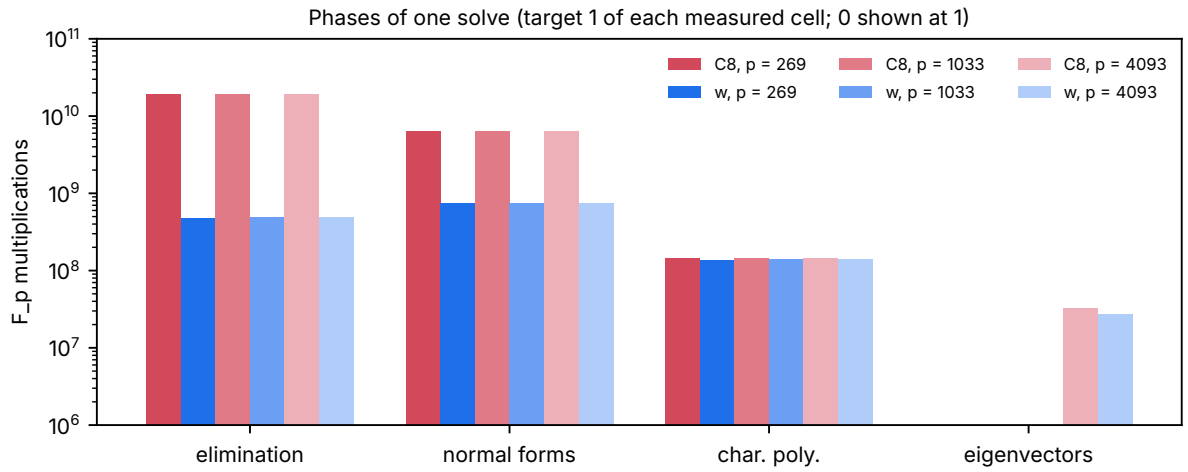


Figure 2: Phases of one solve (target 1, seed 1). Elimination falls by a factor of 40 and normal forms by 8. The characteristic polynomial is unchanged, since $D = 512$ in both arms.

Verdict under the registered rule

- **W1 and W2.**
  - W1 held at $p \geq 29$. At $p = 17$ the dimensions were 510–514.
  - W2 is **not met as registered**: one of 102 targets was unsolved. It was a flagged failure shared with C8, not a wrong set. The decision below therefore carries that caveat.
- **W3** held: the medians are flat in p to 2.2 %.
- **Lead 5b: "projection met".**
  - The median ratio to the plain C_4 is 844–863 at all three sizes, inside $[315, 1260]$.
  - It is not "accounting", because the lift-back is negligible.
- **Against C8: "advances on C8".** The paired ratio is at least 7.7 on all 47 paired targets.

The gap between 48 and 630 was the presentation, not the symmetry.

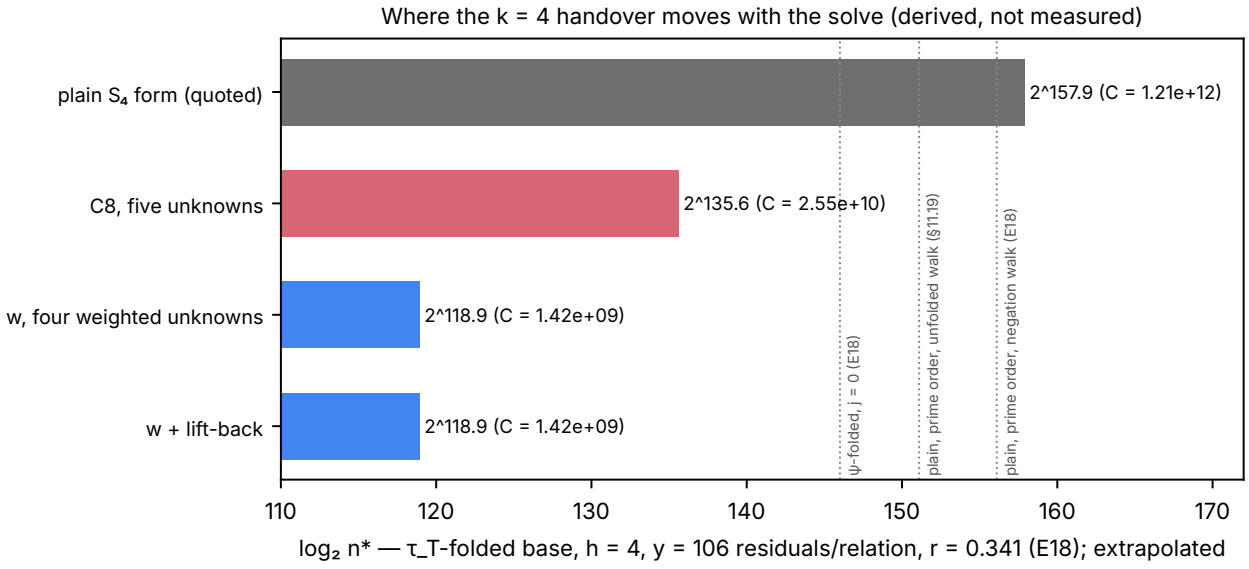


Figure 3: The $k = 4$ handover in E18's τ_T frame. Inputs: $h = 4$, 106 residuals per relation, $r = 0.341$ against the negation walk. The formula is $n^* = 157.9 - 4\log_2(1.213 \cdot 10^{12}/C)$. Derived and extrapolated, not measured.

Handover (derived, extrapolation)

On 2-torsion curves the weighted solve moves the τ_T handover from $2^{157.9}$ to $2^{118.9}$. C8 alone moves it to $2^{135.6}$.

Past the handover the method tends to 0.341 × the negation walk, and r_∞ is unchanged. The figure rests on the exponents $n^{1/4}$ and $n^{1/2}$, and on a C that is flat at $p \leq 4093$. No $k = 4$ run below rho exists.

Review fixes and re-run

The solve now refuses to report "solved" when an eigenspace does not resolve to verified rational points. It reruns one degree higher, up to 28, and otherwise reports unsolved. The handover medians now use one paired set of 24 targets.

All registered cells were re-run with the fixed binary (`raw/v2_*.jsonl`):

- **Outcomes:** no outcome changed on any of the 84 rows.
- **Costs:** one C_w moved, by $-3.3 \cdot 10^5$ ($p = 17$, target 5). Target 4 at $p = 17$ is still unsolved.
- **Handover:** the figures stay at $2^{118.9}$ (w) and $2^{135.6}$ (C8).

Arm w charges the quartic split in the lift-back rather than in C_w. That moves at most $2 \cdot 10^{-4}$ of C_w.

Limits

- The plain arm was not paired on the same targets; it is over the cap.
- No relation phase or linear algebra was run.
- The degenerate targets' cause is unknown.
- Unit tests: 12 of 12 pass (after a test-only fix and the resolution test). The old test planted a zero sum at $p = 13$; the test now uses $p = 29$.
- Recorded C_w values were reproduced exactly from a private-target rebuild after merging main.
- No identity certificate was produced.